Mohd Qasim

+91-6006993954 | qasimallve786@gmail.com | b23ch1027@iitj.ac.in
GitHub | LinkedIn | Portfolio

Academic Qualifications

Indian Institute of Technology, Jodhpur — Sep 2023 – May 2027
Bachelor of Technology in Chemical Engineering — *Jodhpur, Rajasthan*

Jawahar Navodaya Vidyalaya (JNV), Budgam — 2022
Class XII, CBSE — *Budgam, Jammu & Kashmir*

Mutahhary Public High School, Saliskote — 2020
Class X, JKBOSE — *Kargil, Ladakh*

Key Projects

- **DSA Visualizer: Interactive Algorithm Learning Platform** **[Live Site]**
  Team Project, Web Development.
  - Co-developed **AlgoLab**, a learning platform that animates **25+ data structures and algorithms** step by step, including 8 sorting algorithms, 8 graph algorithms (BFS, DFS, Dijkstra, Bellman–Ford, Prim, Kruskal, Topological Sort, Kosaraju SCC), and BST, AVL, B-Tree, Heap, Stack, Queue and Linked List.
  - Migrated the original Ruby on Rails app to **Next.js, React and TypeScript**, splitting it into self-contained feature modules so team members could build and ship visualizers in parallel.
  - Built a snapshot-based step engine in which each algorithm is recorded as pure TypeScript states, enabling play, pause, scrubbing and backward stepping while the executing pseudocode line and live variables are highlighted.
  - Added guided lessons, quizzes, shareable links and runnable code tabs in **Python, C++, Java and JavaScript**, with **Konva.js** canvas engines animating AVL rotations, B-Tree splits and graph traversals.
  - Collaborated through Git feature branches to deploy on **Vercel** with statically pre-rendered, SEO-optimized topic pages and privacy-first PostHog analytics, fuzz-testing each algorithm against reference implementations.
- **Named Entity Recognition (NER)**
  Mentor: Prof. Avinash Sharma , Department of Computer Science Engineering.
  - Built an end-to-end Named Entity Recognition (NER) system using Python and scikit-learn, focusing on entity extraction from unstructured text.
  - Conducted thorough data preprocessing, including handling noisy characters, missing values, and sentence segmentation from raw token-level data.
  - Engineered custom features (e.g., token patterns, casing, word length) and trained a Random Forest classifier for sequence tagging.
  - Evaluated model with NER-specific metrics (precision, recall, F1-score) using the nervaluate library; plotted evaluation results by entity type.
  - Utilized seaborn, matplotlib, and pandas to present results and trends, enhancing model explainability.

Technical Skills

- **Programming Languages:** C, C++, Python, JavaScript, TypeScript
- **Web Development:** Next.js, React, Tailwind CSS, Konva.js, HTML5, CSS3
- **Software and Libraries:** Git, GitHub, Vercel, OOPS, Flask, Aspen Plus
- **Machine Learning Libraries:** Scikit-Learn, PyTorch, TensorFlow, Keras

Relevant Coursework

- **Computer Science:** Introduction to Computer Science, Introduction to Machine Learning
- **Mathematics:** Multivariable and Vector Calculus, Linear Algebra and Differential Equations, Probability, Statistics and Stochastic Processes
- **Core Engineering and Sciences:** Fluid Mechanics, Thermodynamics, Mass Transfer, Material and Energy Balance, Chemical Reaction Engineering, Molecular Engineering, Quantum Chemistry and Spectroscopy